# M M Abid Naziri

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## About Me

PhD Researcher specializing in **Software Testing** for learning-enabled systems. I develop automated testing techniques for Deep Learning Libraries (**PyTorch**, **TensorFlow**) and **Autonomous Driving Systems** in a simulation environment (**CARLA**). I have reported **140** bugs (**103** confirmed) to **PyTorch** and **TensorFlow** to date [list].

## Education

**NC State University**, PhD in Computer Science (5th year) December 2026 (Expected)

**Advisor:** Dr. Marcelo d'Amorim

**Awards:**

- NC State College of Engineering Graduate Merit Award - *June 2025*
- 1st prize in the inaugural NCStateCS Graduate Research Symposium - *April 2026*

## Technologies

**Languages:** Python, R, C#, Delphi, Java, C/C++, JavaScript, TypeScript, Bash

**Technologies:** PyTorch, TensorFlow, CARLA Simulator, Docker, Github Actions, ASP.NET Core, React JS, Vue JS, TestComplete, DUnit

## Experience

**Senior Software Engineer**, Enosis Solutions – Dhaka, Bangladesh Mar 2019 – Jul 2022

- Developed scalable web applications using Vue JS, TypeScript, and ASP.NET Core, including a project management tool deployed to **270** employees and a Welder Management system
- Developed desktop applications DesignCalcs and ProWrite for pressure vessel design, adding features, fixing bugs, and creating unit/regression tests
- Managed CI/CD pipelines, configured testing servers, and ensured smooth system operations
- Operated in an **Agile** development process in a development team of **7** people
- Mentored new recruits, conducted code/design reviews
- Collaborated with clients and team members via **JIRA**

## Research Projects & Publications

**Testing Deep Learning Libraries via Neurosymbolic Constraint Learning** ICSE 2026
***M M Abid Naziri***[*], Shinhae Kim[*], Feiran Qin, Saikat Dutta, Marcelo d'Amorim
[*]Equal contribution
**Venue:** IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

**Misbehavior Forecasting for Focused Autonomous Driving Systems Testing** ICSE 2026
***M M Abid Naziri***, Stefano Carlo Lambertenghi, Andrea Stocco, Marcelo d'Amorim
**Venue:** IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

**A Comprehensive Study of Bugs from Deep Learning Library Fuzzers** *Submitted*
***M M Abid Naziri***, Manideepika Reddy Myaka, Lingjun Liu, Saikat Dutta, Marcelo d'Amorim

**How Effective Is Coverage-Guided Fuzzing to Test Deep Learning Library APIs?** ICST 2026
Feiran Qin, ***M M Abid Naziri***, Hengyu Ai, Saikat Dutta, Marcelo d'Amorim
**Venue:** IEEE International Conference on Software Testing, Verification and Validation (ICST 2026) [pdf]
**Award:** IEEE Computer Society TCSE Best Student Paper Award

**BugsInDLLs: A Database of Reproducible Bugs in Deep Learning Libraries to Enable Systematic Evaluation of Testing Techniques** ISSTA 2025
***M M Abid Naziri***, Aman Kumar Singh, Benjamin Wu, Feiran (Alex) Qin, Saikat Dutta, Marcelo d'Amorim
**Venue:** International Symposium on Software Testing and Analysis (ISSTA 2025: Tool Demos) [pdf]

**Improving Deep Learning Library Testing with Machine Learning** AST 2026
Facundo Molina, ***M M Abid Naziri***, Feiran (Alex) Qin, Alessandra Gorla, Marcelo d'Amorim
**Venue:** ACM/IEEE International Conference on Automation of Software Test (AST 2026) [pdf]